Kunal Pai

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Education

Ph.D., Computer Science, University of California, Los Angeles — Enrolled: September 2026

M.S., Computer Science, University of California, Davis (GPA: **4.0/4.0**) — June 2026

Relevant Skills

Languages: Python, C++, C, JavaScript, Java

ML/AI: TensorFlow, PyTorch, scikit-learn, LLMs, Prompt Engineering, Ollama, Hugging Face, Multi-agent Systems

Web/Data: React, Next.js, Django, Flask, MongoDB, pandas, NumPy, Matplotlib

Tools: Git, Docker, Unix/Linux, Jupyter, LLVM, Clang, gem5

Research & Professional Experience

Graduate Student Researcher, *DavSec Lab @ UC Davis* — April 2025 – Present

- Built an automated pipeline for C-to-Rust transpilation using LLMs, with 5 prompt variations, targeting secure systems migration
- Identified Halstead vocabulary as the strongest metric for predicting translation difficulty
- Validated lightweight semantic augmentations (e.g., filename context) that improved functional accuracy by 5%
- Benchmarked state-of-the-art LLMs across 746 C/C++ programs, achieving 70.2% functional accuracy with best prompt design

Graduate Student Researcher, *DECAL Lab @ UC Davis* — September 2022 – December 2024

- Developed a 4,500-sample dataset for pairwise code-documentation alignment from 200 open-source Python projects, enabling future research in software maintenance
- Engineered a pipeline for measuring calibration and correctness of large language models for code repair, using Defects4J
- Collaborated in validating efficacy of semantic augmentation of language model prompts for code summarization using precision and recall metrics like ROUGE and METEOR

Publications (Selected)

VATS: Exploiting Implicit Authority in Error-Path Injection via Systematic Mutation [Spotlight], Patel, H. & Pai, K., *Second Workshop on Agents in the Wild: Safety, Security, and Beyond, ICML 2026*

CoDocBench: A Dataset for Code-Documentation Alignment in Software Maintenance, Pai, K., Devanbu, P. & Ahmed, T., *International Conference on Mining Software Repositories (MSR) 2025: Data and Tool Showcase Track*

Calibration and Correctness of Language Models for Code, Spiess, C., Gros, D., Pai, K.S., et. al., *International Conference on Software Engineering (ICSE) 2025*

Automatic semantic augmentation of language model prompts (for code summarization), Ahmed, T., Pai, K.S., Devanbu, P. & Barr, E.T., *International Conference on Software Engineering (ICSE) 2024*

Project Experience

NAAMSE: Neural Adversarial Agent Mutation-based Security Evaluator — Nov 2025 – Present

Research Project — *Python, LLMs, Evolutionary Algorithms*

- Won 2nd place (Agent Safety) at the UC Berkeley RDI AgentBeats Competition; framework infrastructure was subsequently forked by Mozilla's 0din team.
- Designed and implemented a clustering engine to identify semantic attack vectors in LLM-generated adversarial prompts.
- Engineered an attack pipeline that iteratively generates, evaluates, and refines adversarial prompts against target models.
- Benchmarked multiple frontier LLMs within the framework to validate attack effectiveness and refine scoring metrics.

SuperNOVA: Superconducting Graph Accelerator in gem5 — January 2024 – March 2026

Research Project, DArchR Lab @ UC Davis — *C++, Python, gem5, Hardware Simulation*

- Modeling a superconducting graph accelerator in gem5, targeting sparse workloads under cryogenic conditions
- Integrating energy and latency estimates from superconducting literature, and RTL simulations, to improve realism
- Mentoring undergraduate researchers on modeling, benchmarking, and research writing; one co-authored a ModSim 2024 poster